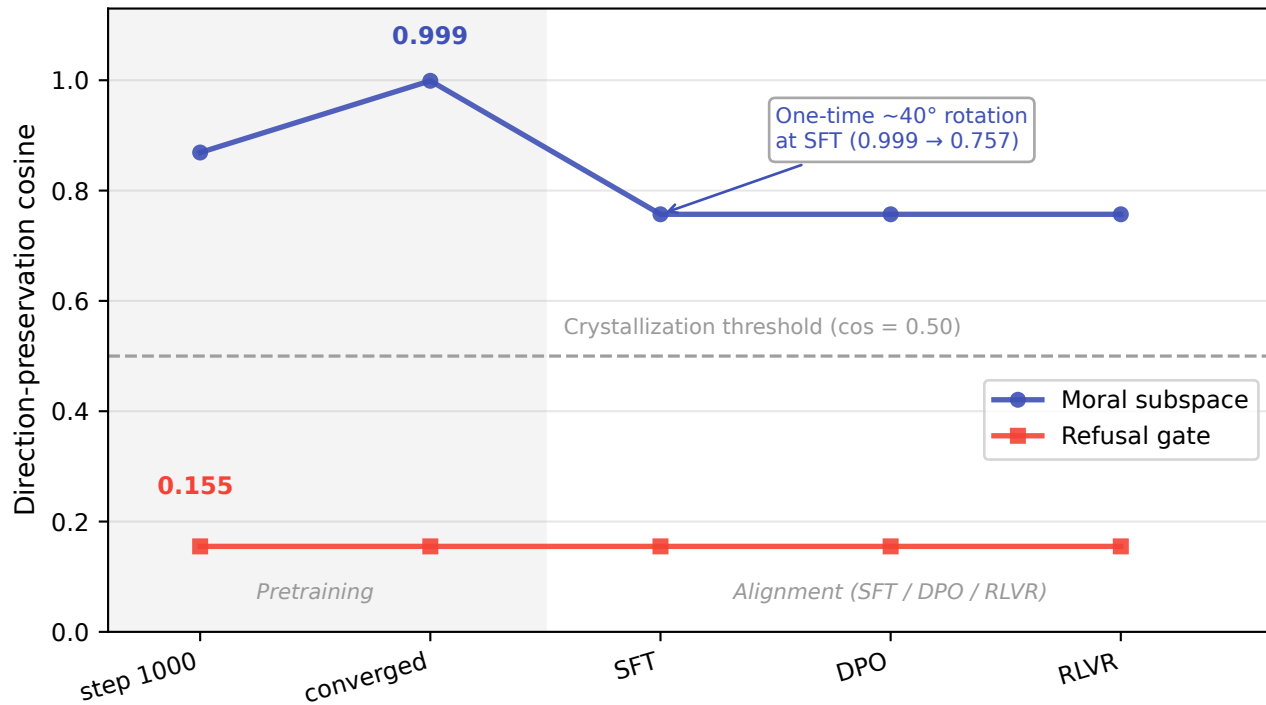


Comprehension crystallizes in pretraining; the refusal gate does not (OLMo-3-7B)

(a) Direction-preservation cosine across pretraining and alignment stages



(b) Precursor crystallizes; alignment to the gate stays flat

